Uganda Christian University | BSc Computer Science

Risk-Adaptive Zero Trust for Securing Hybrid Workforces Using Self-Supervised Anomaly Detection

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Introduction / Problem

Hybrid workforces (remote + on-premise users) expose enterprise networks to ransomware, insider threats, lateral movement, and zero-day exploits.

- Legacy VPNs provide only perimeter-based access — no traffic inspection.
- Zero Trust Architecture (ZTA) improves access control but cannot detect novel, unseen attack patterns.
- Self-supervised ML can detect zero-day anomalies without labelled attack data.

Research Gap

"No existing work pairs Zero Trust Architecture with a self-supervised (label-free) anomaly detector in a comparative, hybrid workforce study."

Gambo & Almulhem (2026): ZTA reviewed — no zero-day ML. Bakhshi & Ghita (2021): DL anomaly detection — requires attack labels. Abdelhady et al. (2026): ZTA + ML for IoT — no hybrid workforce scope.

System Architecture Pipeline

INPUT: NSL-KDD dataset (125,973 train / 22,543 test)

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PREPROCESS: drop zero-variance | log1p | LabelEncode | MinMax → 40 features

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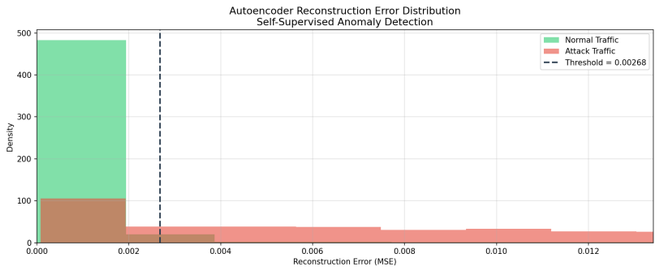
CNN-GRU (Supervised) Conv1D 64→128→256 + GRU 128→64 327K params | **Autoencoder ★ (Self-Supervised) Encoder 40→128→64→16 Decoder 16→64→128→40**

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COMBINED RISK SCORE (0–1) → Flask API → ZTA Policy Engine

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DECISION: ALLOW | MONITOR | CHALLENGE MFA | BLOCK



Reconstruction error: normal vs attack traffic (19× separation)

Key Results — 4-Way Comparison

All systems evaluated on the same 22,543 NSL-KDD test samples.

System	Acc	F1	FPR	Zero-Day
Legacy VPN	0.666	0.652	0.180	No
ZTA Only	0.769	0.768	0.100	No
ZTA+CNN-GRU	0.795	0.789	0.045	No
ZTA+Autoencoder	**0.844**	**0.848**	0.050	**YES ✓**

ZTA+Autoencoder: best accuracy, best F1, only zero-day system.

Per-Class Detection Rates

Class	CNN-GRU	Autoencoder	Winner
DoS	0.817	**0.911**	**AE ★**
Probe	0.571	**0.882**	**AE ★**
R2L	**0.394**	0.287	**CNN**
U2R	0.358	**0.687**	**AE ★**

AE = Autoencoder. R2L is the autoencoder's documented weakness.

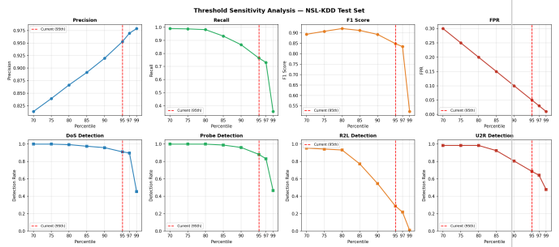
Threshold Sensitivity

Threshold sweep: 70th–99th percentile of normal reconstruction errors.

Pct	Prec	Recall	F1	FPR	Note
70th	0.813	0.990	0.893	0.300	high recall
80th ★	0.866	0.981	**0.920**	0.200	optimal F1
95th	0.953	0.764	0.848	0.050	current
99th	0.979	0.357	0.523	0.010	high prec

Deployment Recommendation

80th pct → F1=0.920 — use for incident-response / high-recall mode. 95th pct → F1=0.848, FPR=0.050 — use for steady-state low-alert environments.



F1/FPR sweep across threshold percentiles (70th–99th)

Zero-Day Detection Capability

Leave-one-class-out experiment: each attack class held out from threshold reconstruction error.

Class	Recall	Sep. Ratio	
DoS	**91.1%**	~22×	High
Probe	**88.2%**	~27.8×	Unu
U2R	**68.7%**	~24.1×	Privi
R2L	**28.7%**	**~3.7×**	Resc

R2L (remote-to-local) connections share byte counts, duration, and service pattern reconstruction error → documented weakness.

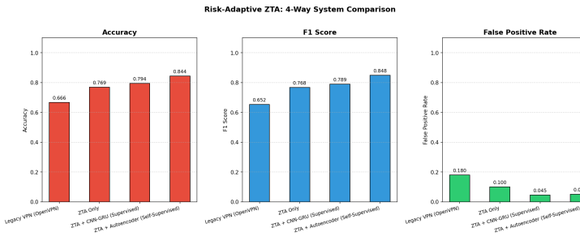
Conclusions

- **Best accuracy (0.844) and F1 (0.848)** of all four systems.
- **Only system with zero-day detection** — self-supervised, no attack labels nee
- **~9× faster inference** than CNN-GRU (0.066 ms vs 0.589 ms/sample) — suitab
- **Honest limitation:** R2L recall 28.7% — those connections resemble normal tra
- **Self-supervised anomaly detection** is a viable, practical complement to super

References

[1] Gambo & Almulhem (2026). Zero Trust Architecture: A Systematic Literature Revie 34(1).
[2] Bakhshi & Ghita (2021). Anomaly Detection in Encrypted Internet Traffic Using H *Comm. Networks.*
[3] Abdelhady et al. (2026). Hybrid ML Anomaly Detection and Lightweight ZTA 18387–18407.

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4-way system comparison: Accuracy, F1, and FPR